

R475: Arch identity isolation and tent rate

TFPT RH programme

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Claim boundary. This note isolates one Mathlib gap and seals a fixed- f quadrature rate. It does not prove compact-window positivity or the Riemann hypothesis. The rate $\text{err}(k, f) \rightarrow 0$ is a fixed- f convergence statement, not a claim about infinitely many k in the cone.

1 A — F1(ii) isolated, not closed

Mathlib v4.29.1 still lists “Prove Gauss’ integral representation” as a TODO on `Digamma.lean`. It supplies `Complex.digamma = logDeriv Gamma`, the recurrence, and the values at $0, 1, 1/2$. There is no `Real.digamma` and no Gauss/Mellin identification.

The r464 exact identity

`ArchGaussMellinDigammaIdentity:`
 $\forall f \exists a_0 \forall a \geq a_0 \forall m \geq \text{meshExp } f,$
`archRead( $a, m, f$ ) = weilArchSide  $f$`

cannot close as stated. Lean now proves

`productionArchDelta\_tendsto\_atTop:`
at each fixed m , $\Delta(a, m) = \log a/(m+1) \rightarrow +\infty$ as $a \rightarrow \infty$.

The inner quantifier therefore asks for exact tent/integral equality on meshes that become arbitrarily coarse. r473 already exhibits a mesh-and-onset-compatible witness with a strictly positive arch tent error. The historical `sorry` is retained (the `FREQ` extraction still consumes the exact form). The missing Mathlib brick is the named Prop `GaussDigammaIntegralRepresentation` — not a second `sorry`.

`weilArchSide` is no longer opaque. It is the concrete u -space pairing

$$A(g) = C_b g(0) + \int_0^b w(u) (e^{-3u/2} g(0) - g(u)) du,$$

$$w(u) = 2e^{-u/2}/(1 - e^{-2u}), \quad C_b = -\gamma - \log \pi - \log(1 - e^{-2b}).$$

Status A: IDENTITY ISOLATED. Census $8 \rightarrow 8$ (honest: exact equality is the wrong quantifier; Gauss is a Mathlib TODO).

2 B — $O(\Delta^2)$ rate at fixed f

Along the selected sequence $\Delta_k = 2^{-\lfloor \sqrt{k} \rfloor} \log 2$, the tent error of the r470 witness is

k	Δ_k	err_{arch}	err/Δ^2
5–8	0.1732867951	0.119793278323	3.9893
9–12	0.0866433976	0.028876619985	3.8466

Halving Δ divides the error by ≈ 4.15 . The rate is $O(\Delta_k^2)$ with witness constant $C \leq 4$.

Lean names

`SelectedArchErrorQuadraticRate`:

$\forall f \exists k_0 \forall k \geq k_0, \text{err}(k, f) \leq \text{archRateConst } f \cdot \Delta_k^2,$

and proves

`selectedArchError\_tendsto\_zero\_of\_rate`:

the named rate plus `selectedDelta\_tendsto\_zero` imply $\text{err}(k, f) \rightarrow 0$ as $k \rightarrow \infty$ at each fixed f .

This is the allowed fixed- f convergence. It is not positivity (that would need the cone for infinitely many k).

The r473 composition is now quantitative: $\text{weilForm}(f) \geq -2\text{err}(k, f)$ with $\text{err}(k, f) \leq C(f)\Delta_k^2 \rightarrow 0$ at fixed f . `SelectedPolynomialApproximatesGrid` is thereby a rate statement, not only an existential residual.

3 Updated joint map

Edge	Status	Content
$A_{\text{cap}} \geq 0 \Rightarrow \text{poly reads} \geq 0$	proved (r473)	<code>selectedACapPsdImpliesPolynomialReads</code>
$\text{onset: fullRead} - \text{weilForm} = \text{err}$	proved (r473)	<code>fullRead_weilForm_gap_eq_arch</code>
weilArchSide concrete u -space	defined (r475)	no longer opaque
$\Delta(a, m) \rightarrow \infty$ at fixed m	proved (r475)	<code>productionArchDelta_tendsto_atTop</code>
$\text{err} \leq C(f)\Delta_k^2$	named (r475)	<code>SelectedArchErrorQuadraticRate</code>
$\text{err}(k, f) \rightarrow 0$ at fixed f	proved of named	<code>selectedArchError_tendsto_zero_of_rate</code>
Gauss integral	named, Mathlib TODO	<code>GaussDigammaIntegralRepresentation</code>
$\text{exact archRead} = \text{weilArchSide}$	sorry retained	wrong quantifier; <code>FREQ</code> consumer

4 Kill table

Candidate	Status	Reason
Close exact F1(ii) this round	killed	$\Delta \rightarrow \infty$ at fixed m ; r473 error > 0 .
Current Mathlib has Gauss	killed	explicit TODO on <code>Digamma.lean</code> .
New <code>sorry</code> for Gauss	killed	named Prop; census stays 8.
Rate is $O(\Delta)$ only	killed	err/Δ^2 stable at 3.85–3.99.
Rate $\Rightarrow$ positivity	killed	anti-list; cone for infinitely many k not claimed.
R475 proves RH	killed	quadrature rate at fixed f .

5 Census

Lean `sorry` census remains 8: three in `Canonical.lean`, the retained exact-identity `sorry` in `Elementwise.lean`, three external arrows in `ExternalBridges.lean`, and one source theorem in `Source.lean`. No new `sorry`. Smoke 9/9, SPEC_SHA e8a49c5895d3d382.